

Maximum Likelihood

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Maximum Likelihood Estimation

- Some simple examples of the most likely parameter value
- Estimating multiple parameters
- Estimator Bias and Mean Squared Error
- The Score, Information, and Minimum Variance Bound
- The Graphical Method
- First look at Likelihood Ratio and Wald test

Intro, reminder of big picture

Maximum Likelihood Estimation is arguably the most important concept for Machine Learning.

Parameter estimation in theory is sold as trying to find the value of a mean, or a rate, or similar.

In practical data science problems, a parameter might be a class with a few possible labels, or a binary yes/no. Choosing the parameter value which is the most likely one is unsurprisingly, always the best choice!

Parameter Estimation

Reminder of some **Model Parameters**:

$K \sim \text{Bernoulli}(p)$ The parameter p is the probability of success
`bernoulli(p=p)`.

$K \sim \text{Binom}(n, p)$ Two parameters, n is the number of Bernoulli trials
`binom(p=p, n=n)`.

$X \sim \text{Uniform}(a, b)$ Two parameters a, b are the bounds of the support
`uniform(loc=a, scale=b-a)`.

$X \sim \text{Expon}(\lambda)$ The parameter λ is rate
`expon(scale=1/λ)`.

$X \sim \text{Norm}(\mu, \sigma)$ Two parameters μ is the mean, and σ is the standard deviation.
`norm(loc=mu, scale=sigma)`.

Estimating Binomial p

According to the data, which value of p is most probable?

Back to tossing a coin! We find k heads in n tosses. Let's Estimate the probability of a tossed coin landing heads up, p .

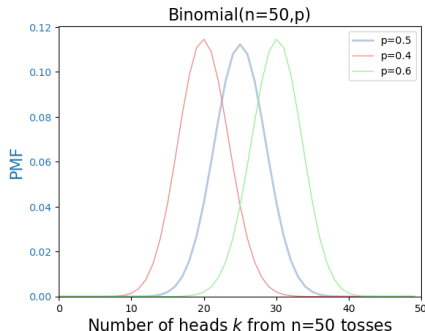
Binomial PMF: $p_K(k) = C_k^n p^k (1 - p)^{n-k}$ Note that $p_K(k)$ is the PMF: the probability of k heads.

We can write this PMF as a conditional Probability:

$$P(k|p) = C_k^n p^k (1 - p)^{n-k}$$

We might as well remind ourselves of the form of $C_k^n = \frac{n!}{k!(n-k)!}$. I am happy to note that this does not depend on the parameter p (it is just a number).

Visual Reminder of Binomial PMF



These plots show the Binomial PMF under different assumptions for our parameter p . The data we measure is k . As we can see, a given measure of k can be consistent with a range of potential p . We want to know, given k , what is the most likely p ?

Estimating Binomial p

The most likely p is our **Maximum Likelihood Estimate** (MLE) for p : $\hat{p}$.

$$p_K(k) = P(k|p) = C_k^n p^k (1-p)^{n-k}$$

I toss the coin $n = 50$ times and observe $k = 27$ heads.

$$\mathcal{L}_p = P(k = 27|p) = C_k^n p^{27} (1-p)^{23} \quad (\text{Note } N = 1, \text{ so } \mathcal{L}_p = \prod_i p_K = p_K)$$

To maximise the Likelihood, we take the derivative **with respect to p** and set it to zero, solving for $\hat{p}$.

$$\frac{d}{dp} (C_k^n p^{27} (1-p)^{23}) = C_k^n (27 p^{26} (1-p)^{23} - p^{27} 23 (1-p)^{22}) = 0$$

We cancel factors of p and $1-p$ to find ... $\hat{p} = \frac{27}{50}$. Not very surprising :)

Log Likelihood

Logs are very useful because eg $\ln(abc) = \ln a + \ln b + \ln c$

Consider a simple machine learning problem where there is a flat probability of $p_K(k) = 0.5 f(\theta)$ and $N = 5 \times 10^3$ data measurements.

The Likelihood is $\mathcal{L}_\theta = \prod_i^N p_K(k)$:

$$\mathcal{L}_\theta = \prod_i^N 0.5 f(\theta) = (0.5 f(\theta))^{5 \times 10^3} \text{ uh-oh!}$$

```
0.5**(5e3) # 0.0
```

```
0.5**(1e3) # 9.332636185032189e-302
```

The Likelihood is not appropriate for numerical computations unless the dataset is really small. And we like big datasets.

Log Likelihood

Logs are very useful because eg $\ln(abc) = \ln a + \ln b + \ln c$

Consider a simple machine learning problem where there is a flat probability of $p_K(k) = 0.5 f(\theta)$ and $N = 5 \times 10^3$ data measurements.

The Log Likelihood is:

$$\ln \mathcal{L}_\theta = \sum_i^N 0.5 f(\theta) = (5 \times 10^3)(0.5 f(\theta))$$

0.5*(5e3) # 2500

The numerical manageability of logs is not the only reason for using them, but is a compelling one. ✓

Log Likelihood for Coin Toss

Reminder: we toss a coin 50 times, see 27 heads. This is a single Binomial probability measurement ($N = 1$), so $\mathcal{L}_\theta = p_k$.

$$\ln \mathcal{L}_p = \ln p_K = \ln \left(C_{27}^{50} p^{27} (1-p)^{23} \right) = \ln \left(C_{27}^{50} \right) + 27 \ln(p) + 23 \ln(1-p)$$

$$\frac{d}{dp} \ln \mathcal{L}_p = 0 + \frac{27}{p} - \frac{23}{(1-p)} = 0 \quad * \text{ chain rule}$$

$\hat{p} = \frac{27}{50}$ as before. We don't lose any information by using the log.

* Chain rule: $f(g) = \ln g$ and $g(\theta) = \mathcal{L}_\theta$, so $\frac{\partial f(g(\theta))}{\partial \theta} = \frac{\partial \ln g}{\partial g} \frac{\partial \mathcal{L}_\theta}{\partial \theta} = \frac{1}{g} \frac{\partial \mathcal{L}_\theta}{\partial \theta} = \frac{1}{\mathcal{L}_\theta} \frac{\partial \mathcal{L}_\theta}{\partial \theta}$

Most Likely Taxi Waiting Time

To estimate the parameter for a continuous RV, we use the PDF rather than the PMF in the binomial example.

The Exponential PDF is $f_T(t; \lambda) = \lambda e^{-\lambda t}$ We use this to model the waiting time t when the rate is λ , with expected value $1/\lambda$.

Data: three days in a row I get a taxi. My waiting times are 15, 18, 11 minutes.

We want to estimate λ from these three data points. Let's write down the Likelihood:

The Likelihood is $\mathcal{L}_\theta = \prod_i^N p_K(k)$:

$$\mathcal{L}_\lambda = (\lambda e^{-\lambda t_{(1)}})(\lambda e^{-\lambda t_{(2)}})(\lambda e^{-\lambda t_{(3)}}) = \lambda^3 e^{-\lambda(t_{(1)}+t_{(2)}+t_{(3)})}$$

Most Likely Taxi Waiting Time

Data: three days in a row I go to wait for a taxi. My waiting times are 15, 18, 11 minutes.

$$\mathcal{L}_{\lambda} = \lambda^3 e^{-\lambda(t_{(1)}+t_{(2)}+t_{(3)})} = \lambda^3 e^{-44\lambda}$$

$$\ln \mathcal{L}_{\lambda} = \ln(\lambda^3) - 44\lambda = 3 \ln \lambda - 44\lambda$$

$$\frac{d}{d\lambda} \ln \mathcal{L}_{\lambda} = \frac{3}{\lambda} - 44$$

$$\hat{\lambda} = \frac{3}{44}$$

Notice that the Estimate for the rate $\hat{\lambda}$ is the reciprocal of the mean waiting time.

Reminder: Log of Gaussian PDF

$$f_Y(y; \theta) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(y-\mu)^2}{\sigma^2}}$$

$$\begin{aligned} \ln f_Y(y; \theta) &= \ln \left[\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(y-\mu)^2}{\sigma^2}} \right] \\ &= \ln \left[\frac{1}{\sigma\sqrt{2\pi}} \right] + \ln \left[e^{-\frac{1}{2} \frac{(y-\mu)^2}{\sigma^2}} \right] \\ &= \ln \left[\frac{1}{\sigma\sqrt{2\pi}} \right] - \frac{1}{2} \frac{(y-\mu)^2}{\sigma^2} \\ &= -\ln \left[\sigma\sqrt{2\pi} \right] - \frac{1}{2} \frac{(y-\mu)^2}{\sigma^2} \\ &= -\ln \sigma - \ln \sqrt{2\pi} - \frac{1}{2} \frac{(y-\mu)^2}{\sigma^2} \end{aligned}$$

log rules : $\ln ab = \ln a + \ln b$

$$\ln e^a = a$$

$$\ln \frac{1}{a} = -\ln a$$

$$-\ln ab = -\ln a - \ln b$$

We have shown

$$\ln f_Y(y; \theta) = -\ln \sigma - \ln \sqrt{2\pi} - \frac{1}{2} \frac{(y-\mu)^2}{\sigma^2}$$

Log of Gaussian Likelihood

The Gaussian distribution has two parameters: μ and σ .

The PDF for a single measurement y_i : $f_Y(y_i | \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(y_i - \mu)^2}{\sigma^2}}$ *

* A measurement represents a range of values, so the PDF for y_i is not zero.

For N measurements of a single RV, the Likelihood is:

$$\mathcal{L}_\theta = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^N e^{-\frac{1}{2} \sum_i \frac{(y_i - \mu)^2}{\sigma^2}}$$

The Log Likelihood is then:

$$\ln \mathcal{L}_\theta = -N \ln \sigma - N \ln \sqrt{2\pi} - \frac{1}{2} \sum_i \frac{(y_i - \mu)^2}{\sigma^2} *$$

* We used another log rule, $\ln a^b = b \ln a$.

Estimation of more than one parameter

$$\ln \mathcal{L}_{\theta} = -N \ln \sigma - N \ln \sqrt{2\pi} - \frac{1}{2} \sum_i^N \frac{(y_i - \mu)^2}{\sigma^2}$$

To find the **Maximum Likelihood Estimate (MLE)**, $\hat{\mu}$, we take the partial derivative with respect to μ , then set it to zero.

$$\frac{\partial}{\partial \mu} \ln \mathcal{L}_{\mu} = 0 - 0 + \sum_i^N \frac{(y_i - \mu)}{\sigma^2} = 0$$

Non-trivial* solution: $\sum_i^N (y_i - \mu) = 0$ when $\sum_i^N y_i = N\mu$

Our MLE for $\hat{\mu} = \frac{1}{N} \sum_i^N y_i$: this is sample mean, $\bar{y}$.

Math meaning of Non-trivial: 'some terms are not zero'.

Estimation of more than one parameter

$$\ln \mathcal{L}_\theta = -N \ln \sigma - N \ln \sqrt{2\pi} - \frac{1}{2} \sum_i^N \frac{(y_i - \mu)^2}{\sigma^2}$$

To find the **Maximum Likelihood Estimate (MLE)** $\hat{\sigma}$, we take the partial derivative with respect to σ , then set it to zero:

$$\frac{\partial}{\partial \sigma} \ln \mathcal{L}_\sigma = -\frac{N}{\sigma} - 0 + \sum_i^N \frac{(y_i - \mu)^2}{\sigma^3} = 0$$

$$\sum_i^N \frac{(y_i - \mu)^2}{\sigma^3} = \frac{N}{\sigma} \text{ when } \sigma^2 = \frac{1}{N} \sum_i^N (y_i - \mu)^2.$$

If we substitute $\mu \rightarrow \hat{\mu} = \frac{1}{N} \sum_i^N y_i = \bar{y}$, we find

$$\hat{\sigma}^2 = \frac{1}{N} \sum_i^N (y_i - \bar{y})^2: \text{ this is the (biased) sample variance, } V[y].$$

Why is the most likely variance biased?

The MLE relies on the data. Unless the dataset includes all possible data, there is likely to be bias in our statistics*.

Previous slide: If we substitute $\mu \rightarrow \hat{\mu} = \frac{1}{N} \sum_i y_i = \bar{y}$, we find

In this step, we are using the sample mean in place of the true mean.

Making this substitution gives us $\hat{\sigma}^2 = \frac{1}{N} \sum_i (y_i - \bar{y})^2$

The MLE is only unbiased in the asymptotic limit ($N \rightarrow \infty$).

* The MLE for variance is not always biased, eg $K \sim \text{Poisson}(\lambda)$ has $E[K] = V[K] = \lambda$, so variance is unbiased.

Mean Square Error (MSE)

Recall the bias of an estimator is $b = E[\hat{\theta}] - \theta$: the difference between the expected value of the estimate $\hat{\theta}$ and the true value of the parameter θ .

The Mean Square Error for an estimate is $MSE = E[(\hat{\theta} - \theta)^2]$

$$\begin{aligned}
 E[(\hat{\theta} - \theta)^2] &= E[(\hat{\theta} - \mu + \mu - \theta)^2] && \text{Add and subtract } \mu = E[\hat{\theta}] \\
 &= E[(\hat{\theta} - \mu)^2] + E[(\mu - \theta)^2] + 2E[(\hat{\theta} - \mu)(\mu - \theta)] && E(\text{sum}) = \text{sum}(E) \\
 &= V[\hat{\theta}] + E[\mu^2 + \theta^2 - 2\mu\theta] + 2\hat{\theta}\mu - 2\hat{\theta}\theta - 2\mu^2 + 2\mu\theta \\
 &= V[\hat{\theta}] + E[\theta^2] - E[\mu^2] + 2E[\hat{\theta}\mu] - 2E[\hat{\theta}\theta] && E(\text{sum}) = \text{sum}(E) \\
 &= V[\hat{\theta}] + E[\theta^2] - \mu^2 + 2\mu^2 - 2\mu E[\theta] && E[E[\hat{\theta}]] = E[\mu] = \mu \\
 &= V[\hat{\theta}] + \theta^2 + \mu^2 - 2\mu\theta && E[\theta] = \theta \\
 &= V[\hat{\theta}] + (\theta - \mu)^2 \\
 &= V[\hat{\theta}] + b^2
 \end{aligned}$$

We have shown $MSE = V[\hat{\theta}] + b^2$

Best Estimator: Smallest MSE

$$MSE = V[\hat{\theta}] + b^2$$

When choosing an Estimator (eg Mean or Median) for a parameter of the model we are using to describe our data, minimising the *MSE* is the goal. There can be a play-off between minimising the bias and the variance of an estimator.

Recall that we found that the sample variance $V[x_N] = \frac{1}{N} \sum_i (x_i - \bar{x})^2$ is a biased estimator for the True Variance:

$$\hat{V}[x_N] = \frac{N-1}{N} V[X] \quad \Leftrightarrow \quad \hat{\theta} = \frac{N-1}{N} \theta; \quad \Leftrightarrow \quad \text{Estimate} = \frac{N-1}{N} \text{Truth}$$

The Estimator Bias is:

$$b = -\frac{V[X]}{N} \quad \Leftrightarrow \quad E[\hat{\theta}] - \theta = E\left[\frac{N-1}{N} V[X]\right] - V[X] = \left(\frac{N-1-N}{N}\right) V[X]$$

Example: If $N = 10$, the bias is $-\frac{1}{10} V[X]$, so $b^2 = \frac{1}{100} V^2[X]$

Best Estimator: Smallest MSE

$$MSE = V[\hat{\theta}] + b^2$$

We skipped over the Variance of the Variance Estimator because it is long, but it is in the books and this is the result:

$$V[\hat{\theta}] = \frac{2N-2}{N^2} \theta^2 \quad \text{Full (confusing) notation is: } V[\hat{V}[X_N]] = \frac{2N-2}{N^2} V[X]^2$$

$$\text{For } N = 10, V[\hat{\theta}] = \frac{18}{100} \theta^2. \text{ Notice this is much larger than } b^2 = \frac{1}{100} \theta^2$$

The Mean Squared Error on the biased variance estimator is therefore:

$$MSE = \frac{2N-1}{N^2} V^2[X]$$

$$\text{For } N = 10, MSE = \frac{19}{100} V^2[X]. \text{ The bias is not a big contributor to the MSE.}$$

Best Estimator: Smallest MSE

$$MSE = V[\hat{\theta}] + b^2$$

The alternative $\hat{\theta} = \hat{V}[X_{N-1}] = \frac{1}{N-1} \sum_i^N (x_i - \bar{x})^2$ is unbiased, $b = 0$.

The variance on this is: $V[\hat{\theta}] = \frac{2}{N-1} V^2[X]$

For $N = 10$, $V[\hat{\theta}] = \frac{2}{9} \theta^2$

The MSE is $\frac{2}{N-1} V^2[X]$

For $N = 10$, $MSE = \frac{2}{9} \theta^2$, which is larger than $\frac{19}{100} \theta^2$

The MSE is better (smaller) if we use the biased sample variance as the estimator for the true variance (for all finite N).

Sometimes, the best estimator for the job is the biased estimator.

The Uncertainties on Max Likelihood Estimates

We have estimated the parameters for Binomial, Exponential, Gaussian. But we do not yet have uncertainties on these estimates.

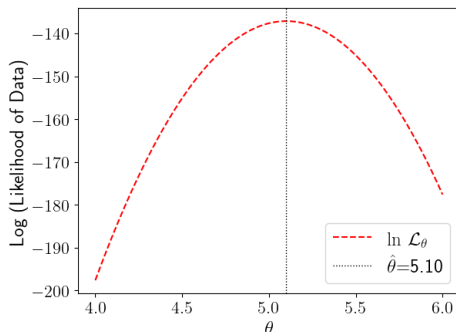
It is not always possible to calculate the uncertainty $\sigma_{\hat{\theta}}$ directly, but we have options:

- 1 Use Monte Carlo (Brute Force). Calculate the Estimate Z times, and use the standard deviation of the Z estimates.
- 2 Use the **Information** to determine the **Minimum Variance Bound**, which we will look at briefly.
- 3 Use the '**Graphical Method**', which we will also look at briefly.

Anatomy of a Log Likelihood Distribution

At the maximum, the first derivative (gradient) is zero, and the second derivative is negative.

The maximum likelihood estimate $\hat{\theta}$ is the most likely value for θ , given the data.



The Score

The first derivative of the Log Likelihood with respect to the parameter(s) of interest is: $\frac{\partial \ln \mathcal{L}_\theta}{\partial \theta}$ *. This is known as **The Score, s** .

*For a vector of parameters $\theta = [\theta_{(1)}, \dots, \theta_{(Z)}]$, $s = \nabla \ln \mathcal{L}$.

The score is the slope (gradient) of the Log Likelihood distribution. **The closer the score is to zero, the closer the parameter is to its maximum likelihood value.**

The Score is $s = \frac{\partial \ln \mathcal{L}_\theta}{\partial \theta} = \frac{1}{\mathcal{L}_\theta} \frac{\partial \mathcal{L}_\theta}{\partial \theta}$ *

* Chain rule: $f(g) = \ln g$ and $g(\theta) = \mathcal{L}_\theta$, so $\frac{\partial f(g(\theta))}{\partial \theta} = \frac{\partial \ln g}{\partial g} \frac{\partial \mathcal{L}_\theta}{\partial \theta} = \frac{1}{g} \frac{\partial \mathcal{L}_\theta}{\partial \theta} = \frac{1}{\mathcal{L}_\theta} \frac{\partial \mathcal{L}_\theta}{\partial \theta}$

The Score in terms of the PDF, f_X

It is useful to write the score in terms of the probability distribution rather than the likelihood, because **probabilities have nice properties like $\int f_X dx = 1$** .

$$\text{Recall } \mathcal{L}_\theta = \prod_i^N f_X(x_i; \theta) \text{ and } \ln \mathcal{L}_\theta = \sum_i^N \ln f_X(x_i; \theta)$$

$$s = \frac{\partial}{\partial \theta} \ln \mathcal{L}_\theta = \frac{\partial}{\partial \theta} \sum_i^N \ln f$$

Adopting shorthand $f \equiv f_X(x_i; \theta)$ for aesthetic reasons.

$$= \sum_i^N \frac{\partial}{\partial \theta} \ln f = \sum_i^N \frac{1}{f} \frac{\partial f}{\partial \theta}$$

Using chain rule and $\frac{d \ln a}{da} = \frac{1}{a}$

$$= N \frac{1}{N} \sum_i^N \frac{1}{f} \frac{\partial f}{\partial \theta} = N E \left[\frac{1}{f_X} \frac{\partial f_X}{\partial \theta} \right]$$

LLN : $\text{plim}_{N \rightarrow \infty} \bar{x} = \mu$

$$= \frac{N}{f_X} \frac{\partial f_X}{\partial \theta}$$

We have shown

$$s = \frac{N}{f_X} \frac{\partial f_X}{\partial \theta}$$

Expected Value of the Score

The score can be helpfully written in terms of the PDF f_X :

$$s = \frac{N}{f_X} \frac{\partial f_X}{\partial \theta}$$

The Expected Value of the score is zero:

$$\begin{aligned} E[s] &= \int s f_X dx \\ &= \int \frac{N}{f_X} \frac{\partial f_X}{\partial \theta} f_X dx \\ &= \int N \frac{\partial f_X}{\partial \theta} dx \\ &= N \frac{\partial}{\partial \theta} \int f_X dx \\ &= N \frac{\partial}{\partial \theta} 1 = 0 \end{aligned}$$

$$\text{LOTUS : } E[g(x)] = \int g(x)f(x)dx = \int g(x)f(x)dx$$

$$\text{Substituting result from previous slide, } s = \frac{N}{f_X} \frac{\partial f_X}{\partial \theta}$$

Switching $\frac{\partial}{\partial \theta}$ and $\int$ is okay under certain conditions

We have shown $E[s] = 0$

Derivative of Expected Value of the Score

$$s = \frac{N}{f_X} \frac{\partial f_X}{\partial \theta}$$

This is zero, because $E[s] = 0$ but we can gain insight from it:

$$\begin{aligned} \frac{\partial}{\partial \theta} E[s] &= \frac{\partial}{\partial \theta} \int s f_X dx = \int \frac{\partial}{\partial \theta} [s f_X] dx \\ &= \int \left[\frac{\partial s}{\partial \theta} f_X + s \frac{\partial f_X}{\partial \theta} \right] dx \\ &= \int \left[\frac{\partial s}{\partial \theta} f_X + \frac{s^2}{N} f_X \right] dx \\ &= \int \frac{\partial s}{\partial \theta} f_X dx + \int \frac{s^2}{N} f_X dx \\ &= E \left[\frac{\partial s}{\partial \theta} \right] + \frac{1}{N} E[s^2] = 0 \end{aligned}$$

Switcheroo

Product rule $\frac{d}{dx} fg = \frac{df}{dx} g + f \frac{dg}{dx}$

Defn. of score : $\frac{\partial f_X}{\partial \theta} = \frac{s f_X}{N}$

Because we started with zero $\left(\frac{\partial}{\partial \theta} E[s] = \frac{\partial}{\partial \theta} 0 = 0 \right)$

We have shown

$$-N E \left[\frac{\partial s}{\partial \theta} \right] = E[s^2]$$

The Variance of the Score

Recall the definition of The Variance of a RV: $V[X] = E[X^2] - E^2[X]$

The Variance of the score is known as The Information, $\mathcal{J}_\theta$:

$$V[s] = E[s^2] - E^2[s]$$

$$= E[s^2]$$

Because $E[s] = 0$, so $E^2[s] = 0$

$$= -N E \left[\frac{\partial s}{\partial \theta} \right]$$

Substituting result from previous slide, $-N E \left[\frac{\partial s}{\partial \theta} \right] = E[s^2]$

$$= -N E \left[\frac{\partial^2 \ln \mathcal{L}_\theta}{\partial \theta^2} \right]$$

Because $s = \frac{\partial \ln \mathcal{L}_\theta}{\partial \theta}$

We have shown

$$V[s] = -N E \left[\frac{\partial^2 \ln \mathcal{L}_\theta}{\partial \theta^2} \right] \equiv \mathcal{J}_\theta$$

Note that $\mathcal{I}_\theta = -E \left[\frac{\partial^2 \ln \mathcal{L}_\theta}{\partial \theta^2} \right] = \frac{\mathcal{J}_\theta}{N}$ is the 'Expected Information' for a single measurement. $\mathcal{J}_\theta$ is the 'Observed Information'.

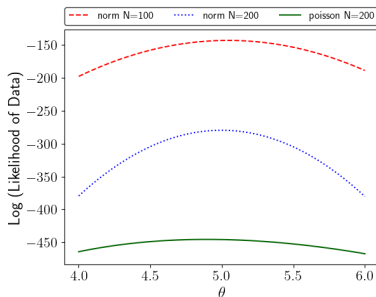
The Information, $\mathcal{J}_\theta$

$$\mathcal{J}_\theta = -N E \left[\frac{\partial^2 \ln \mathcal{L}_\theta}{\partial \theta^2} \right]$$

The Information $\mathcal{J}_\theta$ is so-called because it summarises the information about the parameter that is captured in the data.

A large value of $\mathcal{J}_\theta$ corresponds to a more 'peaky' (more sharply peaked) Log Likelihood distribution, such that it is possible to discriminate between close-by values of θ when estimating the maximum likelihood value, $\hat{\theta}$.

In this plot, the Poisson has larger variance $V[K] = \lambda = 5$ than the Normal $\sigma^2 = 1 \rightarrow$



The Information, $\mathcal{I}_\theta$

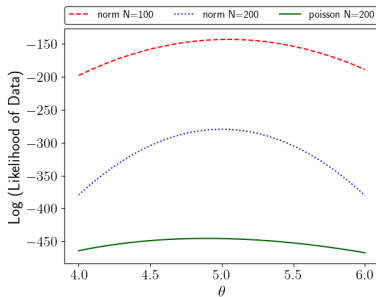
$$\mathcal{I}_\theta = -N E \left[\frac{\partial^2 \ln \mathcal{L}_\theta}{\partial \theta^2} \right]$$

The Information defines the ultimate limit on the accuracy of the estimator, the Minimum Variance Bound, MVB, as we shall see next.

Note that the expectation of the second derivative is its value at the estimate:

$$E \left[\frac{\partial^2 \ln \mathcal{L}_\theta}{\partial \theta^2} \right] \equiv \left. \frac{d^2 \ln \mathcal{L}_\theta}{d\theta^2} \right|_{\hat{\theta}}$$

This is always negative, because the Log Likelihood is **concave**, so the Information is always positive.



The Hessian Matrix

If we were estimating a vector of parameters $\theta = [\theta_1, \dots, \theta_z]$, we would recover an Information Matrix, which is the negative of the **Hessian Matrix** (aka Curvature Matrix):

$$H_{\theta} = \begin{bmatrix} \frac{\partial^2 \ln \mathcal{L}_{\theta}}{\partial \theta_1^2} & \cdots & \frac{\partial^2 \ln \mathcal{L}_{\theta}}{\partial \theta_1 \partial \theta_z} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 \ln \mathcal{L}_{\theta}}{\partial \theta_z \partial \theta_1} & \cdots & \frac{\partial^2 \ln \mathcal{L}_{\theta}}{\partial \theta_z^2} \end{bmatrix}$$

Note that if we were working with the Negative Log Likelihood (convex, with a minimum), the Hessian and the Information Matrix would be identical.

The Information: Normal PDF Example

The Log Likelihood for the Normal PDF yields the following for μ :

$$\ln \mathcal{L}_\theta = -N \ln \sigma - N \ln \sqrt{2\pi} - \frac{1}{2\sigma^2} \sum_i^N (y_i - \mu)^2$$

$$\frac{\partial}{\partial \mu} \ln \mathcal{L}_\theta = \frac{1}{\sigma^2} \sum_i^N (y_i - \mu) \quad \text{First deriv} = 0 \text{ gives us the MLE : } \hat{\mu} = \bar{y}$$

$$\frac{\partial^2}{\partial \mu^2} \ln \mathcal{L}_\theta = \frac{-1}{\sigma^2} \quad \text{Second deriv gives us the information } \mathcal{J}_\theta$$

The Information is:

$$\mathcal{J}_\mu = -N E \left[\frac{\partial^2 \ln \mathcal{L}_\theta}{\partial \theta^2} \right] = -NE \left[\frac{-1}{\sigma^2} \right] = \frac{N}{\sigma^2} \quad N / \text{ True Variance}$$

We have shown

$$\mathcal{J}_\mu = \frac{N}{\sigma^2}$$

The Information and Minimum Variance Bound

For $Y \sim \text{Norm}(\mu, \sigma)$, the information (for μ) is the inverse of the variance of the estimate $\hat{\mu}$:

$$\mathcal{J}_{\mu} = \frac{N}{\sigma^2} = \frac{N}{V[Y]} = \frac{1}{V[\bar{Y}]} = \frac{1}{V[\hat{\mu}]}$$

Recall variance on the mean is $V[\bar{Y}] = \frac{V[Y]}{N}$

Generally, the Rao-Cramer-Frechet (RCF) Inequality provides a Minimum Variance Bound: the variance on an (unbiased) estimate cannot be smaller than the inverse information.

$$V[\hat{\theta}] \geq \frac{1}{\mathcal{J}_{\theta}}$$

If the estimator is the Minimum Variance Unbiased Estimator (MVUE, ie the best estimator!) then $V[\hat{\theta}] = \frac{1}{\mathcal{J}_{\theta}}$, as per the example above for Gaussian $\hat{\mu}$.

The Minimum Variance Bound

The **Minimum Variance Bound (MVB)** of an Estimator is the ultimate limit of its accuracy. If the Estimator is biased ($b = E[\hat{\theta}] - \theta \neq 0$), then we must use the **full version of the RCF inequality**:

$$V[\hat{\theta}] \geq \left(1 + \frac{db}{d\theta}\right)^2 / \mathcal{J}_{\theta}$$

The lower bound on the **Mean Squared Error** ($MSE = V[\hat{\theta}] + b^2$) is then:

$$MSE \geq \left(1 + \frac{db}{d\theta}\right)^2 / \mathcal{J}_{\theta} + b^2$$

As we have seen, the MSE for a biased estimator can be smaller than the MSE for an unbiased one, if the biased variance is smaller than the unbiased variance.

Taylor Series Expansion of $f(x)$ about a

Recall the Taylor series expansion for a function $f(x)$ about the value $x = a$:

$$f(x) = \sum_0^{\infty} \frac{f^n(a)}{n!} (x - a)^n$$

Let's write it out in full for a few terms:

$$f(x) = f(a) + \left. \frac{df}{dx} \right|_a (x - a) + \frac{1}{2!} \left. \frac{d^2f}{dx^2} \right|_a (x - a)^2 + \frac{1}{3!} \left. \frac{d^3f}{dx^3} \right|_a (x - a)^3 + \dots$$

Taylor Series Expansion of $\ln \mathcal{L}_\theta$ about $\hat{\theta}$

$$f(x) = f(a) + \left. \frac{df}{dx} \right|_a (x - a) + \frac{1}{2!} \left. \frac{d^2 f}{dx^2} \right|_a (x - a)^2 + \frac{1}{3!} \left. \frac{d^3 f}{dx^3} \right|_a (x - a)^3 + \dots$$

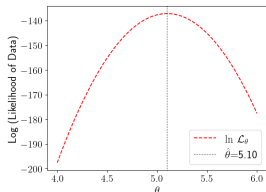
Let's write this down for the Log Likelihood function, $f(x) \rightarrow \ln \mathcal{L}_\theta$, expanding about the MLE: $a \rightarrow \hat{\theta}$:

$$\ln \mathcal{L}_\theta = \ln \mathcal{L}_{\hat{\theta}} + \left. \frac{d \ln \mathcal{L}_\theta}{d\theta} \right|_{\hat{\theta}} (\theta - \hat{\theta}) + \frac{1}{2!} \left. \frac{d^2 \ln \mathcal{L}_\theta}{d\theta^2} \right|_{\hat{\theta}} (\theta - \hat{\theta})^2 + \dots$$

Why have I stopped at the quadratic term? Lazy?

Taylor Series Expansion of $\ln \mathcal{L}_\theta$ about $\hat{\theta}$

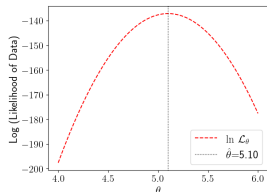
$$\ln \mathcal{L}_\theta = \ln \mathcal{L}_{\hat{\theta}} + \left. \frac{d \ln \mathcal{L}_\theta}{d\theta} \right|_{\hat{\theta}} (\theta - \hat{\theta}) + \frac{1}{2!} \left. \frac{d^2 \ln \mathcal{L}_\theta}{d\theta^2} \right|_{\hat{\theta}} (\theta - \hat{\theta})^2$$



$\ln \mathcal{L}_{\hat{\theta}}$ is the maximum value of the Log Likelihood, at $\theta = \hat{\theta}$

Taylor Series Expansion of $\ln \mathcal{L}_\theta$ about $\hat{\theta}$

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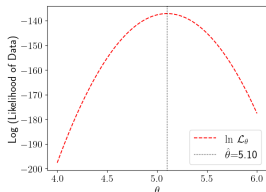


$\ln \mathcal{L}_{\hat{\theta}}$ is the maximum value of the Log Likelihood, at $\theta = \hat{\theta}$

$$\left. \frac{d \ln \mathcal{L}_\theta}{d\theta} \right|_{\hat{\theta}} = 0 : \text{ the first derivative (score) is zero by definition at } \theta = \hat{\theta}.$$

Taylor Series Expansion of $\ln \mathcal{L}_\theta$ about $\hat{\theta}$

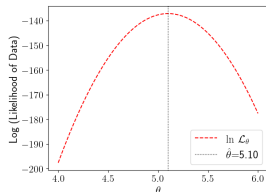
$$\ln \mathcal{L}_\theta = \ln \mathcal{L}_{\hat{\theta}} + \cancel{\frac{d \ln \mathcal{L}_\theta}{d\theta} \bigg|_{\hat{\theta}} (\theta - \hat{\theta})} + \frac{1}{2!} \frac{d^2 \ln \mathcal{L}_\theta}{d\theta^2} \bigg|_{\hat{\theta}} (\theta - \hat{\theta})^2$$



$\ln \mathcal{L}_{\hat{\theta}}$ is the maximum value of the Log Likelihood, at $\theta = \hat{\theta}$

$$\frac{d \ln \mathcal{L}_\theta}{d\theta} \bigg|_{\hat{\theta}} = 0 : \text{the first derivative (score) is zero by definition at } \theta = \hat{\theta}.$$

Taylor Series Expansion of $\ln \mathcal{L}_\theta$ about $\hat{\theta}$



$$\ln \mathcal{L}_\theta = \ln \mathcal{L}_{\hat{\theta}} + \frac{1}{2!} \left. \frac{d^2 \ln \mathcal{L}_\theta}{d\theta^2} \right|_{\hat{\theta}} (\theta - \hat{\theta})^2$$

$\ln \mathcal{L}_{\hat{\theta}}$ is the maximum value of the Log Likelihood, at $\theta = \hat{\theta}$

$\left. \frac{d \ln \mathcal{L}_\theta}{d\theta} \right|_{\hat{\theta}} = 0$: the first derivative (score) is zero by definition at $\theta = \hat{\theta}$.

$\left. \frac{d^2 \ln \mathcal{L}_\theta}{d\theta^2} \right|_{\hat{\theta}}$: the second derivative is negative (therefore not zero) for a maximum.

Evaluated at $\theta = \hat{\theta}$, this is the expected value.

$$\left. \frac{d^2 \ln \mathcal{L}_\theta}{d\theta^2} \right|_{\hat{\theta}} = E \left[\frac{d^2 \ln \mathcal{L}_\theta}{d\theta^2} \right] = -\frac{1}{V[\hat{\theta}]}$$

Graphical Method

$$\ln \mathcal{L}_\theta = \ln \mathcal{L}_{\hat{\theta}} + \frac{1}{2} \frac{d^2 \ln \mathcal{L}_\theta}{d\theta^2} \bigg|_{\hat{\theta}} (\theta - \hat{\theta})^2 = \ln \mathcal{L}_{\hat{\theta}} - \frac{1}{2} \frac{(\theta - \hat{\theta})^2}{V[\hat{\theta}]}$$

We can now ask the question: what is the value of $\ln \mathcal{L}_\theta$ at $\theta = \hat{\theta} \pm \sigma_\theta$?

$$\ln \mathcal{L}_{\hat{\theta} \pm \sigma_\theta} = \ln \mathcal{L}_{\hat{\theta}} - \frac{1}{2} \frac{(\hat{\theta} \pm \sigma_\theta - \hat{\theta})^2}{V[\hat{\theta}]} = \ln \mathcal{L}_{\hat{\theta}} - \frac{1}{2} \frac{\sigma_\theta^2}{V[\hat{\theta}]} = \ln \mathcal{L}_{\hat{\theta}} - \frac{1}{2}$$

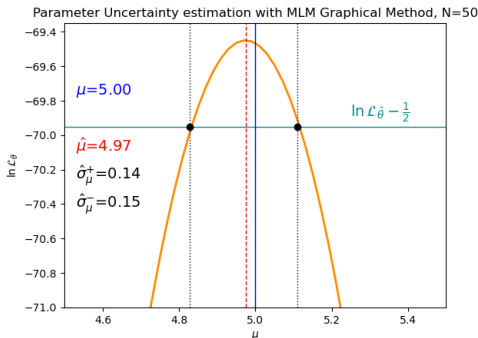
We have shown

$$\ln \mathcal{L}_{\hat{\theta} \pm \sigma_\theta} = \ln \mathcal{L}_{\hat{\theta}} - \frac{1}{2}$$

Extension to $n\sigma$ with $n > 1$ is just: $\ln \mathcal{L}_{\hat{\theta} \pm n\sigma_\theta} = \ln \mathcal{L}_{\hat{\theta}} - \frac{n^2}{2}$

Graphical Method: Visually

$$\ln \mathcal{L}_{\hat{\theta} \pm n\sigma_{\theta}} = \ln \mathcal{L}_{\hat{\theta}} - \frac{n^2}{2}$$



The graphical method is an astonishingly simple way to estimate the uncertainty on a maximum likelihood estimate. No calculus needed! The simplicity of this method is thanks to the simplicity of the shape of the log likelihood - **take care, as this is not always the case!**

MLE as an Interval Estimator

Note that the value of the Likelihood on its own is $\sim$ meaningless, but a **ratio** allows us to compare two models.

$$\text{Likelihood Ratio: } \lambda = \frac{\mathcal{L}_{\theta}}{\mathcal{L}_{\hat{\theta}}}$$

The numerator, $\mathcal{L}_{\theta}$, is for example the Likelihood for the parameter value that would correspond to our favourite theory being the correct one. Or, it could be the Likelihood for some boring null hypothesis. Either way, it cannot by definition be larger than the denominator (the maximum likelihood!), so $0 \leq \lambda \leq 1$.

The test statistic commonly used for the likelihood ratio is

$$t_{\theta} = -2 \ln \frac{\mathcal{L}_{\theta}}{\mathcal{L}_{\hat{\theta}}}$$

MLE as an Interval Estimator

Likelihood Ratio Test Statistic:

$$t_{\theta} = -2 \ln \lambda = -2 \ln \frac{\mathcal{L}_{\theta}}{\mathcal{L}_{\hat{\theta}}} = -2(\ln \mathcal{L}_{\theta} - \ln \mathcal{L}_{\hat{\theta}})$$

Recall our conclusion from the Taylor expansion:

$$\ln \mathcal{L}_{\theta} = \ln \mathcal{L}_{\hat{\theta}} - \frac{1}{2} \frac{(\theta - \hat{\theta})^2}{V[\hat{\theta}]} \rightarrow -2(\ln \mathcal{L}_{\theta} - \ln \mathcal{L}_{\hat{\theta}}) = \frac{(\theta - \hat{\theta})^2}{V[\hat{\theta}]}$$

This (approximate) likelihood ratio test statistic is called the **Wald Test** and allows for the easy extraction of Confidence Intervals, as per the graphical method, in certain simple situations (PDF approximately Normal) with lots of data (large N).

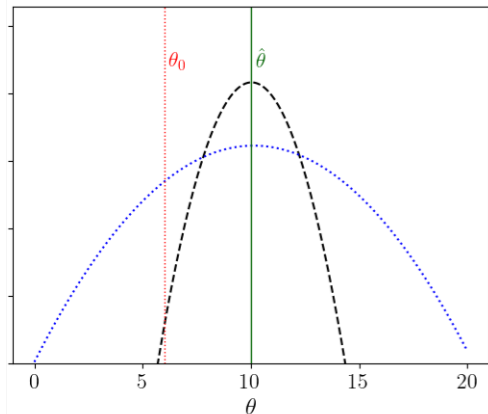
The Wald test statistic:

$$W = \frac{(\theta - \hat{\theta})^2}{V_N[\hat{\theta}]}$$

Wald Test, Visually

Wald:

$$W = \frac{(\theta - \hat{\theta})^2}{V_N[\hat{\theta}]}$$



Thanks For Your Time

Fin.